DE RUEDA CONSTRUCTION

MASTER WEBSITE & CONSTRUCTION ERP SPECIFICATION

Complete rebuild from the beginning

Public website • Private ERP • Client records and email • Sites & projects • Adjustments • HR • Attendance • Payroll • PayMongo • Canteen GCash • Materials • Finance • Reports • Security • Deployment

DOCUMENT PURPOSE	One authoritative, consolidated implementation specification
CLIENT TERMINOLOGY	The person or organization commissioning construction work is the Client
PAYMENT DECISION	PayMongo for supported payment use cases; capabilities must be verified
SITE CANTEEN	GCash destination is configurable per site with controlled access and audit history
BUILD STANDARD	npm only • no Bun • npm run build must pass before release

Important: this document specifies intended behavior. It does not certify that features are already built, connected to live providers, or deployed.

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Operating rule: Keep features simple, secure, auditable, and honest about implementation status.

1. Purpose, scope, and truthfulness rules

This document is the complete, rebuilt-from-the-beginning specification for the De Rueda Construction public website and private browser-based operations system. It consolidates the previously discussed requirements into one coherent reference, including Client records and email addresses, site/project adjustment workflows, PayMongo, canteen GCash configuration by site, HR, attendance, weekly payroll, materials, finance, reports, security, visual design, and deployment.

Product boundaries

- The public company website and private ERP are separate areas. Public visitors must never be able to browse private operational records.
- The ERP is an online/browser system unless a separate native application is explicitly approved. It must work well on desktop and mobile browsers.
- All construction sites use the same core modules. Access and data visibility may be restricted by assigned role, site, project, and action.
- Every requirement in this document is a target specification. A listed feature is not proof that it has been implemented, connected to a live service, or tested in production.
- Clearly label mockups, sample records, test-mode integrations, manual workflows, and production-verified features. Never present simulated records or statuses as real.

Implementation principles

- Use one authoritative saved record for each employee, Client, site/project, payment, inventory movement, and payroll run.
- Preserve audit history. Archive/inactivate records instead of deleting records needed for legal, financial, or operational history.
- Use stable unique IDs; names and email addresses are attributes, not substitutes for primary record IDs.
- Validate inputs and permissions on the server, not only in the browser interface.
- Deliver a small, reliable feature set first. Do not add unnecessary complexity, duplicate modules, or decorative controls that do nothing.

2. Brand, navigation, and visual design

The interface should look like one coherent, professional construction-company system rather than a collection of unrelated screens.

Brand standard

- Use the preferred glossy green “de RUEDA” wordmark with an architectural/metallic emblem and the subtitle “CONSTRUCTION”.
- Use green, dark blue, light blue, white, and black as the core palette. Pink glitter is allowed only as a restrained accent.
- Do not revert to the rejected old yellow/gold hardhat logo.
- Use clear typography, high contrast, consistent spacing, restrained shadows, readable tables, and responsive layouts.

Public navigation

- Home
- About Us
- Services

- Projects / Portfolio
- Careers / Hiring
- Business Inquiry
- Contact
- Private System Login

Private ERP navigation

- Executive Dashboard
- Clients
- Sites & Projects
- Project Adjustments
- Employees & HR
- Attendance / DTR
- Weekly Payroll
- Payments & Transactions
- Canteen
- Materials & Suppliers
- Expenses
- Documents
- Reports
- Approvals
- Activity / Audit History
- Settings & Access Control

Interaction rules

- Every visible button, link, menu, filter, card, and form control must work as labeled or be clearly marked unavailable.
- Provide loading, empty, validation, success, failure, offline/network error, and permission-denied states.
- Use confirmation dialogs for high-impact actions; avoid confirmations for routine harmless actions.
- Show breadcrumbs or clear back navigation in nested site/project screens.
- Use consistent status labels and explain what each status means.

3. Public company website

The public website presents the company, attracts legitimate inquiries and applicants, and directs authorized staff to the private ERP.

Pages and content

- Home: company introduction, approved brand, primary services, selected public projects, contact call-to-action, and hiring/business inquiry links.
- About Us: approved company profile, values, capabilities, and factual information supplied by the company.

- Services: construction services with clear descriptions; do not claim licenses, certifications, projects, or guarantees that have not been verified.
- Projects / Portfolio: only approved public project summaries and photos. Never expose private client details or restricted project documents.
- Careers / Hiring: applicant form with role interest, contact details, optional permitted attachment, consent/privacy notice, and submission confirmation.
- Business Inquiry: route prospective Clients, suppliers, subcontractors, and partners to an authorized reviewer.
- Contact: official public contact information and an accessible inquiry method.
- System Login: entry point to the private ERP; no private dashboard content is rendered before authentication.

Public form behavior

- Validate required fields on both client and server; show useful errors without exposing system internals.
- Apply spam protection, rate limits, safe file-type/size limits, and secure upload handling.
- Store submissions in the approved database or a specifically configured company workflow. Do not assume Google Drive is connected until tested.
- Track application/inquiry states such as New, Under Review, Contacted, Shortlisted, Closed, and Archived.
- Only authorized staff may review submissions. Public users must not be able to enumerate or download other submissions.

4. Client records and Client email requirements

In the company system, the owner/customer commissioning work is referred to as the Client. Each project must link to a Client record, and the Client’s email address must be a real structured field—not free-text notes or an email copied into a report manually.

Client profile fields

- Client ID: unique, stable internal identifier.
- Client type: individual, company, organization, or other approved category.
- Client/company name and contact person's full name.
- Contact person's position/role, where relevant.
- Primary email address, optional secondary email address, and email verification state.
- Optional phone/mobile number and correspondence address.
- Preferred communication channel and language, if needed.
- Consent/communication preference, record source, created-by user, creation timestamp, last-updated timestamp, and status.
- Internal notes with restricted visibility; do not mix private internal notes into Client-facing messages.

Client email validation and maintenance

- Normalize leading/trailing spaces and validate basic email syntax; syntax validation alone does not prove the mailbox belongs to the Client.
- Support an optional verification process using a time-limited confirmation link or a staff-verified confirmation. Store verification status and timestamp.
- Allow authorized staff to correct an email address, require a reason for significant changes, and retain the prior value in audit history.
- Warn staff if a project has no usable Client email before attempting an email report. Do not silently send to a guessed address.

- Where multiple contacts exist, mark one as the primary recipient and permit project-specific recipients only under controlled permissions.
- Do not treat an email address as a login credential. Client contact data does not automatically create a Client portal account or grant ERP access.

Relationship to sites and projects

- One Client may own or commission multiple projects. Each project must link to the correct Client ID.
- Store project-specific recipient overrides only when needed, with the reason, author, timestamp, and approval where required.
- Display Client name and primary email in the project Overview and Client/Contacts tab, subject to the viewer's permissions.
- Changing a Client's default email must not rewrite old report records. Each sent report retains the recipient address used at the time.
- Prevent duplicate Client records where possible by prompting staff to search existing Client names, emails, and organization details before creating a new record.

Client privacy and access

- Client-facing reports may include progress, dated photos approved for sharing, completed work, upcoming work, schedule notes, and approved communications.
- Never include employee payroll, individual rates, SSS/PhilHealth numbers, private deductions, internal-only notes, restricted supplier pricing, or unrelated Client information.
- Only authorized roles can export Client lists or view sensitive contact fields. Record bulk exports in the audit log.
- Do not expose Client email addresses in public pages, source code, public file links, or unauthenticated APIs.

5. Client communications and report delivery

Email reporting should be reliable, reviewable, and safe. A generated PDF is not considered sent merely because it was created or a button was clicked.

Customer progress reports

- Support milestone progress reports at 25%, 50%, 75%, and 100% completion when those milestones are applicable.
- Support monthly reports between milestones when configured for that project.
- Include project name/ID, reporting period, progress summary, completed work, current activities, next steps, dated approved photos, known delays/issues, and prepared-by details.
- Generate a customer-ready PDF and allow preview before sending.
- Require a human review/approval before the first release of a report and before sending revised reports, unless a later explicit policy approves automation.
- Show the recipient email clearly in the confirmation screen; staff must be able to cancel before sending.
- Record report version, reporting period, selected photos, approver, recipient, send attempt, timestamp, provider/message reference if available, and delivery state.
- Support retry after failure without creating a duplicate report or misleading the user about delivery.
- Provide a manual download/copy workflow if outbound email delivery has not been configured.

Email states

- Draft: not ready for delivery.
- Ready for Review: generated and awaiting approval.
- Approved: approved to send, but not yet submitted to the email service.
- Sending: a send request is in progress.
- Sent / Accepted: email provider accepted the request; this does not guarantee that a human read it.
- Delivered: only if the provider returns a reliable delivery event.
- Failed / Bounced: delivery failed or the destination was rejected.
- Cancelled: user stopped the send before submission.
- Needs Review: timeout or ambiguous provider response; do not automatically claim success or failure.

Email provider and security

- Use an approved email service or company mailbox integration; no provider is assumed to be connected by this specification.
- Keep credentials server-side in environment variables or a secrets manager. Never place email API keys in browser code or Git.
- Use appropriate authentication for the sending domain, including SPF/DKIM/DMARC where supported and configured by the domain owner.
- Limit recipients, prevent unauthorized bulk sending, and log the acting user and reason for report delivery.
- Minimize personal information in email subject lines and notification previews. Use secure links for sensitive documents rather than public attachments where appropriate.

6. Authentication, roles, permissions, and CEO controls

Authentication establishes identity; authorization determines exactly what that identity may do. Both must be enforced by the backend for every protected operation.

Account security

- Support authorized account invitations/creation, login, logout, password recovery, session expiry, and session revocation.
- Use secure password hashing. Never store plaintext passwords or hard-code a default CEO password.
- Support multi-factor authentication and passkeys/biometrics where technically available, with a secure recovery method.
- CEO first-time activation must verify identity, establish credentials and recovery, and create an audit record.
- Log sign-in, sign-out, failed attempts, recovery events, role changes, access denials, and privileged actions.

Role model

- CEO / Owner: company-wide oversight, final approvals, delegation, role administration, and audit access.
- Admin 1 / HR: employee records, onboarding, attendance review, payroll preparation, and explicitly permitted HR workflows.
- Admin 2 / Operations or Payroll: assigned operational functions and sites only.
- Site Supervisor / Site Head: assigned-site attendance, daily reports, material receipt/usage, progress, and operational exceptions.
- Finance / Approver: approved expense, payment, and reconciliation functions.
- Read-only / Auditor: permitted views and reports without edit or approval rights unless explicitly granted.

Permission and delegation rules

- Scope permissions by role, site, project, data category, and action: view, create, edit, approve, export, archive, restore, or configure.
- CEO delegation must record the delegate, exact permissions, site/project scope, effective dates, reason, and revocation status.
- Delegation must not silently grant all CEO privileges. A delegated action must be traceable to the acting user and delegation.
- Restrict SSS/PhilHealth identifiers, payroll, personal documents, payment settings, Client exports, and sensitive financial reports.
- High-impact approvals should not be self-approved by the requester unless a documented exception policy explicitly allows it.

7. Executive dashboard

The dashboard is an overview of persisted company data. Each metric should link to the records behind it and show when it was last updated.

- Employee headcount, new hires, inactive employees, and site distribution.
- Today's attendance: present, absent, late, missing time-out, and pending corrections.
- Weekly payroll: draft, under review, approved, locked, and verified payment state where supported.
- Projects by Draft, Ongoing, Paused, Completed, Cancelled/Unfinished, Archived, and Restored status.
- Materials alerts: low stock, reserved quantities, pending deliveries, and site requests.
- Budget versus actual expenses, pending approvals, and transactions needing proof or reconciliation.
- Client progress reports due, awaiting approval, sent, failed, or needing review.
- Project adjustment requests awaiting review and approved adjustments not yet applied.
- Recent audit activity, security events, and unresolved operational exceptions.

Dashboard controls

- Provide date-range, site, project, department, Client, and status filters where relevant.
- Cards open the relevant filtered module instead of being decorative.
- Show last-updated timestamps and distinguish live values from cached values.
- Do not show balances, payout outcomes, or provider status unless backed by verified data.

8. Employee management and HR

Employee record fields

- Unique Employee ID, legal name, job title, employment type, start date, employment status, site assignment, supervisor, and compensation basis.
- Contact and emergency contact details where company policy requires them.
- Employment documents, contract, onboarding checklist, certificate-of-employment history, and effective-dated changes.
- Protected SSS/PhilHealth and other statutory details with access restricted to authorized HR roles.
- Payment destination details only when required by the approved workflow; mask these fields in ordinary views.

Onboarding and changes

- Search for potential duplicate records before creating a new employee; names may repeat, Employee IDs may not.

- Validate required identity, job, rate, contact, and site fields before activating the employee.
- Require reasons for important changes and preserve old/new values, editor, timestamp, and approval if applicable.
- Site transfers use effective dates and preserve historical assignments.
- Termination changes employment status but does not erase attendance, payroll, deductions, or documents.
- Use an Archive/Junk Employee Records area for archived records with controlled restore/reactivation.
- Rehire should link to prior history and avoid duplicate identity records.
- Permanent deletion requires elevated authorization, retention review, and an audit record; use archive by default.
- Generate COE using the approved company template and approved CEO name/title; preview before download or print.

9. Attendance, DTR, and site supervision

Capture and review

- Record date, site, employee ID, exact clock-in and clock-out timestamps, and the source of the entry.
- Support QR capture if implemented and authorized, with manual fallback for permitted users.
- Do not treat a QR scan as valid unless identity, timestamp, and site are saved server-side.
- Keep exact times to the minute; do not silently round times or overwrite original punches.
- Apply approved rules for breaks, payable hours, overtime, and exceptions rather than hard-coding assumptions.
- Record absence, lateness, missed punches, overtime, and corrections with a reason.
- Site Supervisor reviews assigned-site entries; HR reviews exceptions before payroll cutoff.
- Corrections preserve original values, corrected values, actor, timestamp, reason, and approval trail.

DTR and site report outputs

- Search by date, site, employee ID/name, status, and exception type.
- Generate daily/weekly DTR with clock-in/out, breaks, hours, exceptions, reviewer, and correction history.
- Export print-ready PDF and Excel with filters and generated timestamp.
- Provide end-of-day checklist for missing time-outs, absences, site conditions, progress, delays, and workforce concerns.
- Site daily reports include workforce count, completed work, materials received/used, contractor/equipment issues, safety observations, blockers, photos, and follow-ups.

10. Weekly payroll, deductions, and payslips

Calculation requirements

- Select pay period/week and site or employee group.
- Use approved attendance and effective-dated compensation rates.
- Calculate base earnings and configured allowances, overtime, and adjustments according to approved company policy.
- Itemize SSS, PhilHealth, loans, cash advances, canteen balances, and other authorized deductions.
- Show gross pay, each deduction, total deductions, and net pay per employee and payroll batch.
- Flag missing attendance/rates, duplicate lines, unusual changes, negative net pay, and inconsistent totals.

Approval and finalization

- Payroll remains Draft until reviewed and approved.
- Record approver, timestamp, pay period, batch total, and notes.
- Lock or version the approved payroll snapshot so later attendance edits cannot silently change it.
- Use controlled adjustment/reversal workflow for post-approval corrections.
- A payment request is not proof of completed payment; require verified provider status or documented manual evidence.

Payslips and notifications

- Include pay period, earnings, itemized deductions, net pay, and verified payment reference/status only when available.
- Protect individual payslips so employees cannot view other employees' records.
- Trigger payout notifications only after the configured payment confirmation is verified.
- Keep SMS/email content minimal and avoid exposing excessive sensitive details on a lock screen.
- Record notification destination, time, delivery state where available, retries, and failures.

11. PayMongo and payments ledger

PayMongo is the selected payment provider for supported payment flows. The exact products available to the company account must be checked against current official documentation and account permissions before implementation.

Supported scope and limits

- Use PayMongo for supported incoming customer payments and status tracking.
- Do not assume incoming payment collection automatically supports employee payroll payouts or arbitrary outgoing disbursements.
- Use outgoing disbursement features only if the relevant official product/API and company account explicitly support the required use case.
- If a required payout flow is unsupported, implement a clearly labeled authorized manual process rather than a fake integration.
- Use a general “Payments & Transactions” module, not unnecessary provider-specific wallet hubs.

Payment lifecycle and ledger

- Create an internal transaction ID linked to Client, invoice/project, amount, currency, purpose, and actor.
- Create checkout sessions/payment links on the backend using server-side credentials; expose only safe checkout details to the frontend.
- Store provider references, expected amount/currency, creation time, status, and reconciliation state.
- Verify webhook authenticity using PayMongo's current documented method; process events idempotently.
- Handle duplicate webhooks without duplicating ledger entries, receipts, or notifications.
- Mark Paid only after verified confirmation. Distinguish Pending, Failed, Expired, Cancelled, Refunded, and Needs Review where supported.
- Link confirmed payments to the correct Client/invoice/project and reconciliation record.
- A timeout or delayed webhook is ambiguous; use Needs Review or server-side status checking rather than inventing an outcome.

Security and testing

- Keep secret keys server-side in environment variables or a secrets manager; never expose them in frontend bundles or Git history.
- Separate test and live credentials and clearly label test-mode transactions.
- Restrict payment settings, refunds where supported, and manual payment records to authorized roles.
- Test successful payment, decline/failure, cancellation, expiry, duplicate webhook, mismatched amount/currency, delayed webhook, refund where supported, and network timeout.
- Reconcile provider records against the internal ledger before declaring the integration production-ready.

12. Canteen and site-specific GCash configuration

The canteen module has a narrow, clear requirement: the canteen should be able to receive GCash payments, and the displayed/recorded GCash destination must be editable per site by an authorized user. This is separate from PayMongo customer payment collection and must not be represented as a live GCash API integration unless one is officially configured.

Per-site configuration

- Each site can have its own canteen payment configuration, including approved GCash recipient/display name and mobile number or QR reference as permitted by company policy.
- Show the selected site's payment instructions only within that site's canteen context.
- Allow only authorized roles to create or edit the site's GCash details; require confirmation and record who changed what, when, and why.
- Keep prior configuration history so staff can audit which destination was active on a given date.
- Validate required fields and show a clear warning if the site's GCash configuration is missing or inactive.
- Do not expose private credentials, OTPs, PINs, or account passwords in the system. Never ask staff to store them in site settings.

Canteen transaction records

- Record canteen transaction ID, site, employee ID where relevant, transaction date, item/amount, payment method, reference/evidence if provided, recording user, and reconciliation state.
- Do not automatically mark a transaction as received solely because someone typed a reference number. Use the approved verification process.
- Support Pending Verification, Confirmed, Rejected/Invalid, and Needs Review states as appropriate.
- Do not claim automated GCash balance viewing, payout execution, or API verification unless official access has been granted and the feature has been tested.
- Keep canteen balances/deductions linked to the correct employee and payroll period, with approval and correction history.

13. Sites, projects, status, and project profile layout

Project master record

- Unique Project ID/name, linked Client ID, site address, Client contact details, assigned manager/supervisor, budget/contract value, start/planned completion dates, and documents.
- Project status: Draft, Ongoing, Paused, Completed, Cancelled/Unfinished, Archived, and Restored.
- Every status change records actor, date/time, previous status, new status, reason, and notes.
- Link employees, contractors, materials, expenses, progress updates, issues, Client communications, and attachments.
- Show schedule risks, budget use, blockers, outstanding approvals, and data freshness where available.

Project profile tabs

- Overview
- Client & Contacts
- Work Scope
- Schedule & Milestones
- Budget
- Team & Assignments
- Materials
- Expenses
- Documents
- Customer Updates
- Adjustments
- Activity History

Status transitions

- Ongoing → Paused and Paused → Ongoing require a reason and effective date.
- Ongoing → Completed requires completion checks and an authorized approver.
- Active → Cancelled/Unfinished requires a reason, financial/material reconciliation where applicable, and preservation of open obligations.
- Archive hides a project from routine active lists but retains authorized historical access.
- Restore requires permission and validation of any required setup before returning the project to active workflows.

14. Site and project adjustment center

Adjustments are controlled changes to an existing site or project. A change must not silently rewrite approved budgets, schedules, scope, staffing, material quantities, or historical reports.

Adjustment record fields

- Unique Adjustment ID; linked Project ID and Site ID; category; title; detailed reason; requester; created timestamp; requested effective date; evidence/attachments; current state.
- Before value, proposed after value, delta, impact summary, and related records.
- Reviewer/approver, decision timestamp, decision notes, and application timestamp.
- Version number and link to any adjustment that supersedes or corrects this request.

Adjustment categories

- Work scope or work-item changes.
- Budget, contract value, cost allocation, or approved financial limit.
- Schedule, milestone, start date, or completion date.
- Site address/details, operational status, hold, cancellation, or restoration.

- Staffing, employee assignments, supervisor, or contractor assignment.
- Material type, quantity, supplier, delivery, or allocation.
- Expense allocation, customer-facing reporting settings, and Client contact/recipient changes.

Workflow states

- Draft → Submitted → Under Review.
- Under Review → Approved, Rejected, or Returned for Changes.
- Approved → Applied only after the system successfully commits the change.
- Draft or Submitted may be Cancelled by an authorized user according to policy.
- Applied adjustments are not edited in place; create a correction or superseding adjustment.

Impact analysis and approvals

- Before approval, show the current value, proposed value, difference, affected modules, and known downstream consequences.
- Assess impact on budget totals, schedule, staff assignments/attendance, payroll, material stock, expenses, Client communications, and reports.
- Require approval thresholds by change category and amount; configure thresholds with the company rather than inventing values.
- Requester and approver should be different for high-impact changes unless a documented exception is enabled.
- Applying an approved adjustment must be transactional: either all linked updates succeed or none are committed.
- Prevent duplicate application if the user double-clicks or the request is retried.
- Record every state transition, approval, rejection, return, application, and correction in append-only history.

History and reporting

- Keep the original value and effective date so authorized users can view “as-of” historical reports.
- Approved historical payroll, expense, payment, and inventory records must not be silently recalculated by a later project adjustment.
- Allow filters by project, site, category, requester, approver, state, and date.
- Show pending adjustments on the project dashboard and block affected operations where policy requires approval first.

15. Materials, suppliers, contractors, and inventory

Catalog and stock

- Material ID/name, specification, unit, category, default supplier, unit cost, reorder threshold, and active/archive status.
- Track on-hand, reserved, available, and in-transit quantities separately where implemented.
- Use a documented stock valuation method; balances must be derivable from movement history.
- Low-stock alerts link directly to the affected material and site.

Purchases and deliveries

- Requests include site, material, quantity, reason, needed-by date, supplier, and approval state.
- Track purchase/order reference, expected delivery, received/rejected/damaged quantities, cost, receipt/invoice, and receiver.

- Support partial delivery; never mark an entire order received if only part arrived.
- Maintain supplier/contractor contacts, scope, linked project, documents, delivery history, and notes.

Release, use, and adjustments

- Record quantity issued, site, receiver, timestamp, and reference.
- Record consumption, returns, and stock adjustments with reason and author.
- Prevent negative stock unless an explicitly approved and audited override exists.
- Preserve movement history; do not rewrite old stock movements to hide a discrepancy.

16. Expenses, documents, Google Drive, and internal messages

Expenses and cash flow

- Capture vendor, category, amount, date, linked site/project, purpose, receipt, requester, approver, and notes.
- Show budget versus actual, commitments, payroll liabilities, and categorized totals where supported by saved data.
- Require approval at configured thresholds and preserve approval history.
- Distinguish Draft, Pending, Approved, Confirmed Paid, Failed, Refunded/Reversed where applicable, and Reconciled.
- A receipt supports a record but is not, by itself, proof that a payment completed.

Documents and Drive

- Maintain a searchable document index: title, category, linked employee/project/site/Client, owner, access class, upload date, version metadata, and file/link location.
- Google Drive may be used within the company's configured quota and sharing policy; the current planning constraint is a 1 TB allowance, so manage file size and retention.
- Store document links and metadata where appropriate rather than creating uncontrolled duplicate copies.
- Restrict employee, payroll, contract, and finance documents. Avoid public links for sensitive records.
- Do not claim a file is safely backed up or access-controlled unless the actual Drive permissions and integration have been verified.

Internal messages

- Support internal direct/group messages with timestamps and site/project context where implemented.
- Keep sensitive HR/payroll attachments out of general operational groups.
- Route public applications to HR and business inquiries to assigned staff with status, reviewer, follow-up, and archive history.

17. Reports, exports, and audit history

Report catalog

- Executive summary and site dashboard.
- HR roster, onboarding, transfers, archive, and COE records.
- Attendance/DTR, absences, lateness, overtime, exceptions, and corrections.

- Payroll register, deductions, net pay, payslips, approvals, and reconciliation.
- Project status, progress, schedule, budget, Client reports, and adjustment history.
- Materials stock, low stock, purchases, deliveries, releases, usage, and supplier records.
- Finance income/payment records, expenses, outstanding items, refunds/disputes where supported, and reconciliation exceptions.
- Client communication/report delivery history and failed email sends.

Export rules

- Filter by date, site/project, employee, Client, supplier, status, and category where relevant.
- Preview where practical; include report title, period, applied filters, generation timestamp, and page numbers.
- Generate print-ready PDF (letter size where required) and useful Excel (.xlsx) files with headings, currency, and date formats.
- Sensitive HR/payroll fields are excluded by default and require explicit export permission.
- Empty results remain empty; never fabricate rows, values, or charts.

Audit events

- Record actor, action, entity type/ID, timestamp, outcome, and reason/context for sensitive actions.
- Include role changes, employee edits, attendance corrections, payroll approval, payment configuration, expense approvals, Client email changes, report sending, project adjustments, archive/restore, exports, and security events.
- Restrict audit-log modification and access. Never log passwords, OTPs, API secrets, or full payment credentials.

18. Data model and cross-module integrity

Core entities

- User, Role, Permission, Delegation, Session, SecurityEvent.
- Client, ClientContact, ClientCommunicationPreference, Project, Site, Milestone, SiteReport, ProgressReport, ProjectAdjustment.
- Employee, EmploymentHistory, EmployeeDocument, SiteAssignment, CompensationRate.
- AttendanceEvent, AttendanceException, AttendanceApproval, PayrollRun, PayrollLine, Deduction, Payslip, PayrollAdjustment.
- Material, InventoryMovement, PurchaseRequest, PurchaseOrder, Delivery, SiteRelease/Usage.
- Expense, Budget, PaymentRecord, ProviderEvent, ReconciliationRecord, DocumentReference, Message, PublicApplication, BusinessInquiry, NotificationLog, AuditEvent.

Integrity rules

- Employee IDs, Client IDs, project IDs, site IDs, adjustment IDs, and transaction IDs are unique and stable.
- Attendance references a valid employee, date, and site when site-scoped.
- Payroll lines reference the payroll run, employee, compensation basis, attendance source, earnings, and deductions.
- Each project references the correct Client record; sent reports preserve the actual recipient used.
- Payment records have unique internal IDs and provider references when returned; every status transition is traceable.
- Inventory balances derive from movement history and approved adjustments.
- Archived records remain available in authorized historical reports.
- Validate required fields, numeric ranges, state transitions, and cross-record relationships on the server.

Time and currency

- Use a consistent timestamp policy and display Philippine local time where appropriate.
- Preserve exact attendance timestamps and use approved rules for payroll calculations.
- Store money in precise numeric form and display Philippine peso (PHP) formatting consistently.
- Use the same time and currency conventions across attendance, payroll, reports, payment events, and audit logs.

19. Security, privacy, backups, and recovery

Security controls

- Enforce server-side authorization for every private API, report, export, file, and privileged action.
- Use HTTPS, secure sessions/cookies, CSRF protections where applicable, input validation, rate limits, and safe error messages.
- Mask sensitive identifiers and restrict government IDs, payroll, loans, cash advances, personal documents, Client exports, and payment settings.
- Store secrets server-side and rotate exposed credentials; never commit credentials to source control.
- Separate test and production environments, keys, databases, and payment events.

Resilience and backups

- Back up the database and document metadata; define and test a restore procedure.
- Handle timeouts and webhook retries safely. A timeout does not prove success or failure.
- Use idempotency for financial actions and external event processing.
- Keep logs useful without exposing private data.
- When a provider is unavailable, use pending/review states instead of inventing results.
- Define retention/access rules for public applications, former employees, payroll, receipts, contracts, Client communication, and audit events.

Privacy and compliance

- Collect only necessary applicant and Client information and explain its purpose.
- Restrict customer-facing reports to approved information.
- Do not use unrestricted public Drive links for sensitive documents.
- Provide a process to correct inaccurate Client contact information and to handle communications preferences.
- Confirm applicable Philippine privacy and employment obligations with the company's responsible adviser before production launch.

20. UI acceptance and workflow requirements

Critical workflows to demonstrate

- Create Client → add verified/usable email → create project linked to Client → preview progress report → approve → send → review actual send status.

- Change Client email → audit old/new value → verify new address if required → ensure prior sent reports retain their original recipient.
- Create project adjustment → compare before/after values → review impact → approve/reject/return → apply once → verify linked modules and audit history.
- Record attendance → review exception → approve correction → calculate payroll draft → approve/lock payroll → reconcile payment evidence.
- Configure canteen GCash destination for Site A → verify Site B remains unchanged → edit Site A configuration → confirm change history.
- Create PayMongo test payment → receive/verify provider event → prevent duplicate event processing → reconcile ledger → keep test mode visibly distinct.
- Archive project/employee → confirm historical reports remain available to authorized users → restore with appropriate checks.
- Export a report → verify permissions, filters, file formatting, and audit entry.

Definition of done

- Every required screen is connected to real saved records or clearly labeled as not yet implemented.
- Role/site restrictions are tested using accounts with different permissions.
- Invalid inputs, missing Client email, duplicate submissions, retries, and network errors are tested.
- PDF and XLSX exports are checked for readable formatting and correct totals.
- Email delivery is verified through the configured provider; a local preview alone does not count.
- Payment integration is tested in test mode before live credentials are enabled.
- All critical workflows pass on desktop and mobile browser sizes.

21. Secret Recipe: build, deployment, and release gate

These are the standing project rules for website changes and deployment.

- Standardize on npm. Do not use Bun as the package manager for this project.
- Remove conflicting Bun lockfiles (bun.lock or bun.lockb) when standardizing to npm.
- Keep package.json and package-lock.json synchronized. CI should use npm ci for reproducible installs.
- Run npm run build and fix all build errors before declaring the project ready for deployment.
- Confirm the actual framework build output directory. Use dist/ for Cloudflare Pages only when the project is configured to output there.
- Verify the deployment project, branch, commit, build command, environment variables, and output directory.
- Test the deployed HTTPS site on desktop and mobile: routes, forms, authentication, permissions, Client email workflow, project adjustments, reports, and payment test flow.
- Do not claim deployment is successful without checking the deployment result and live site.
- Do not present mock data, simulated payment states, placeholder email sending, or unimplemented buttons as production features.
- Keep test and production PayMongo credentials separate; use server-side secrets and verify webhook handling and idempotency.
- Maintain a concise release note describing what changed, what was tested, what remains manual, and any known limitations.

Release checklist

- Dependencies install cleanly with npm ci.

- npm run build succeeds with no unresolved errors.
- Lockfile is committed and synchronized.
- Correct build output directory is confirmed.
- Environment variables and secrets are configured safely.
- Deployment logs show the intended branch and commit.
- Live HTTPS smoke tests pass on desktop and mobile.
- Critical Client, project adjustment, payroll, canteen, report, and payment flows are tested.
- Backup/restore and role-permission checks are completed.
- Release status is reported truthfully with evidence.

22. Phased implementation order

Phase 1 — foundations

- Establish repository/build baseline, npm-only workflow, database schema, authentication, roles, audit log, and responsive layout.
- Build Client records and email field early because project setup and report delivery depend on it.
- Build Sites & Projects, Client linking, project status history, and site-scoped access.

Phase 2 — daily operations

- Implement employee onboarding, attendance/DTR, supervisor daily reports, materials/inventory, expenses, and document index.
- Implement site-specific canteen GCash configuration with audit history and a clearly defined manual verification workflow.
- Implement project Adjustment Center and impact/approval rules before enabling high-impact changes.

Phase 3 — payroll, reports, and communications

- Implement weekly payroll calculation, review, approval, locking, payslips, and controlled corrections.
- Implement PDF/XLSX reports, Client progress report preview/approval, and email delivery through a configured provider.
- Implement notification logs and failure/retry handling.

Phase 4 — payments and launch

- Integrate PayMongo only for supported company-account capabilities, beginning in test mode.
- Verify webhook authenticity, idempotency, reconciliation, and failure cases.
- Complete security review, backup/restore test, role-based acceptance tests, mobile/desktop checks, and production smoke tests.
- Launch only after critical acceptance tests pass; keep unsupported functions clearly labeled as manual or unavailable.

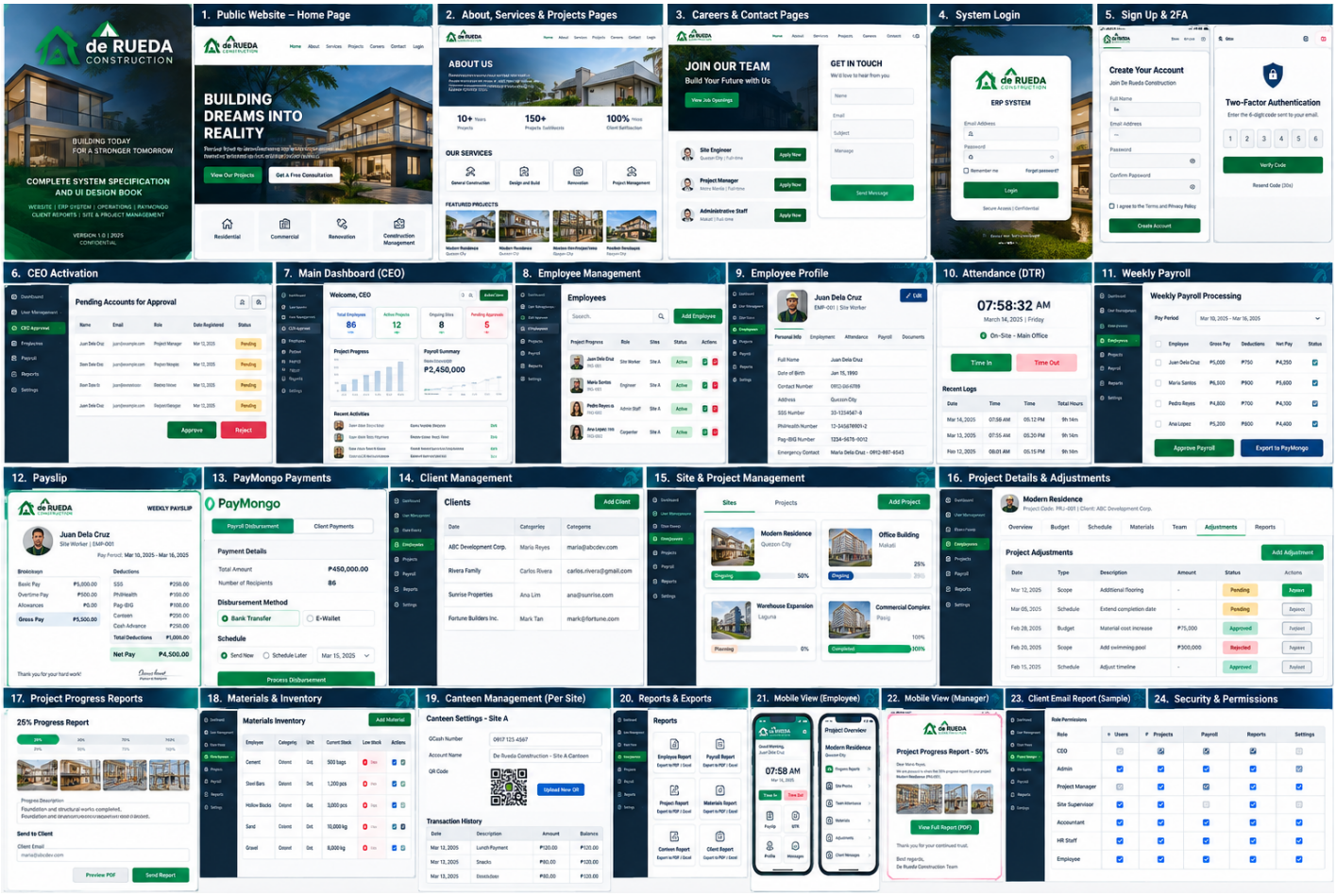
23. Final decision register

- Product form: public website plus private browser-based construction ERP.
- Client terminology: use “Client” for the person/company commissioning a project.
- Client email: first-class structured field on Client/contact records; visible in project contacts; used by the approved report-delivery workflow.

- Payment provider: PayMongo for supported payment use cases; do not assume payroll payouts are supported.
- Canteen: receives GCash according to the approved process, with editable site-specific recipient details and audit history.
- Project adjustments: formal request/review/approval/application process with before/after values and impact analysis.
- Site/project statuses: Draft, Ongoing, Paused, Completed, Cancelled/Unfinished, Archived, Restored, with auditable transitions.
- Reports: PDF and XLSX, with letter-size PDF where required; Client-facing reports must exclude internal HR/payroll data.
- Storage: Google Drive may be used within configured policy and the noted 1 TB allowance; verify actual integration and permissions.
- Deployment: npm only, no Bun, synchronized lockfile, npm run build must pass, verify output directory and live HTTPS deployment.
- Truthfulness: document requirements do not prove implementation. Verify every feature before calling it live.

UI DESIGN REFERENCE APPENDIX

Original generated design overview • Reference only, not proof of live implementation



Screen-by-screen UI inventory

The generated overview contains 24 concept screens. Use the written requirements in the main specification as the source of truth for behavior, permissions, data, and integration status. Labels and sample values inside mockups are illustrative and must be replaced with approved real data.

01 Public Website — Home — Brand hero, services, project highlights, contact and inquiry calls to action.

02 About, Services & Projects — Company profile, service cards, public portfolio, and approved project imagery.

03 Careers & Contact — Hiring openings, applicant form, contact form, and privacy notices.

04 System Login — Private ERP sign-in, validation, recovery, and secure access entry.

05 Account Setup & 2FA — Authorized account onboarding, second factor, recovery, and security guidance.

06 CEO Activation — Pending account review, role approval, and audit trail for first-time owner setup.

07 CEO Dashboard — Company-wide KPIs, projects, payroll summary, approvals, alerts, and recent activity.

08 Employee Management — Searchable employee list, unique Employee ID, onboarding, and permission-aware actions.

09 Employee Profile — Employment data, site assignment, documents, compensation, and restricted statutory data.

10 Attendance / DTR — Exact time punches, site context, recent entries, exceptions, and authorized corrections.

11 Weekly Payroll — Pay period selection, earnings, itemized deductions, exception review, approval, and exports.

12 Payslip — Employee-specific earnings, deductions, net pay, pay period, and verified payment reference if available.

13 PayMongo Payments — Supported payment flow, transaction state, reconciliation, and test/live separation; not an assumed payroll payout UI.

14 Client Management — Client/company name, contact person, email, verification state, related projects, and communication history.

15 Sites & Projects — Site/project list, status, completion indicators, Client association, and project navigation.

16 Project Details & Adjustments — Project tabs, proposed before/after changes, approval state, impact summary, and adjustment history.

17 Project Progress Report — Milestone progress, dated approved photos, summary, PDF preview, recipient check, and send controls.

18 Materials & Inventory — Stock quantities, reorder alerts, deliveries, issue/usage history, and site-scoped inventory.

19 Canteen — Per-Site GCash — Editable recipient/display details per site, QR reference, transaction records, and verification state; never store PIN/OTP.

20 Reports & Exports — PDF/XLSX catalog, filters, preview, permission checks, and generated timestamp.

21 Employee Mobile View — Responsive mobile attendance, payslip access, and employee-appropriate functions.

22 Manager Mobile View — Responsive manager site/project overview, approvals, and assigned-site workflows.

23 Client Email Report — Example progress email, recipient confirmation, report link/attachment policy, and delivery status.

24 Security & Permissions — Role/site permission matrix, delegated access, approval scope, and auditability.

Design interpretation and implementation guardrails

• These screens are conceptual visual references generated for planning. They are not screenshots from the deployed application.

• Some sample text, figures, names, timestamps, and transaction amounts visible in a mockup are illustrative only and must never be treated as company data.

• Use the approved De Rueda green architectural wordmark. Core palette: green, dark blue, light blue, white, and black; pink glitter is only a subtle accent.

• Use PayMongo only for payment capabilities actually enabled for the company account. Do not imply that PayMongo automatically pays employees.

• Canteen GCash details are configurable per site. Do not store GCash PINs, passwords, OTPs, or private login credentials.

• Client email is a structured Client/contact field; each sent report retains the actual recipient, version, approver, send attempt, and delivery status available from the email provider.

• All private data and actions must be protected server-side, not merely hidden in the interface.

• All controls must work or be explicitly labeled unavailable. No fake success messages or simulated live integrations.

• Project adjustment approvals must preserve before/after values, impact analysis, actor, effective date, and versioned audit history.

• Follow the project build rule: npm only, no Bun, synchronized package-lock.json, npm run build must pass, and verify the actual deployment and live HTTPS site.